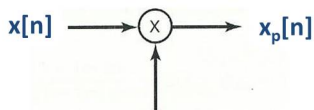


Homework: 8.4, 8.25, 8.29  
Tutorial Problems: 8.34, 8.35, 8.38



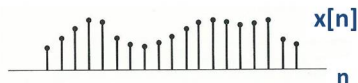
# Sampling on Discrete-Time Signals

- In digital signal processing, we may sample the discrete-time signals
- Example: image, audio, video compression
- Its mathematical model is give below:



$$p[n] = \sum_{k=-\infty}^{+\infty} \delta[n - kN]$$

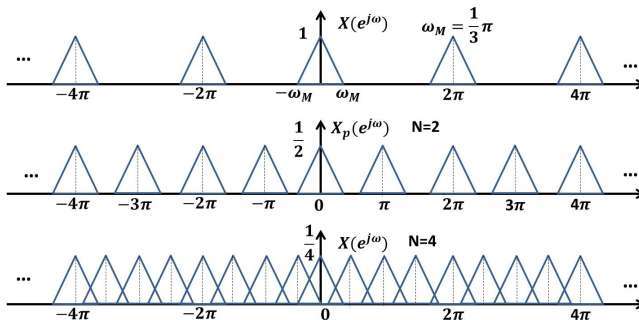
$$x_p[n] = \sum_{k=-\infty}^{+\infty} x[kN] \delta[n - kN]$$



# Frequency Analysis

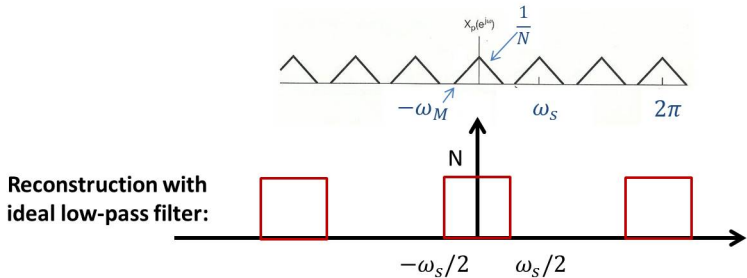
$$P(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s) \quad \text{where } \omega_s = 2\pi/N$$

$$X_p(e^{j\omega}) = \frac{1}{2\pi} \int_{2\pi} P(e^{j\theta}) X(e^{j(\omega-\theta)}) d\theta = \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j(\omega-k\omega_s)})$$



# Reconstruction

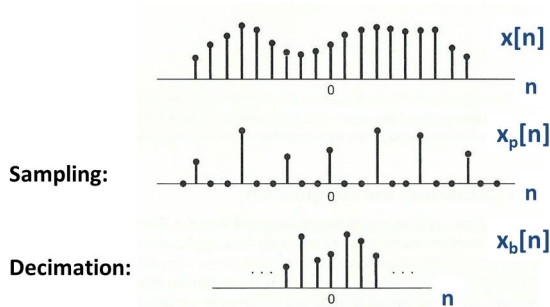
- Perfect reconstruction is applicable when  $\omega_s > 2\omega_M \leftrightarrow N < \frac{\pi}{\omega_M}$



- Aliasing occurs when  $\omega_s < 2\omega_M$

# Decimation

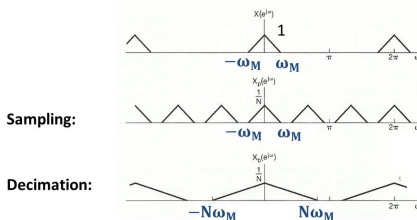
- After sampling, there will be a great amount of redundancy
- **Decimation**: discrete-time sampling + remove zeros



$$x_b[n] = x_p[nN] = x[nN]$$

# Frequency Analysis

$$\begin{aligned}
 X_b(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x_b[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x_p[n]e^{-j\omega n/N} = X_p(e^{j\omega/N}) \\
 &= \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j\omega/N - k\omega_s})
 \end{aligned}$$



**Condition for Perfect Reconstruction:  $N\omega_M < \pi$**

Intuitively, what happens in frequency domain when we do decimation?

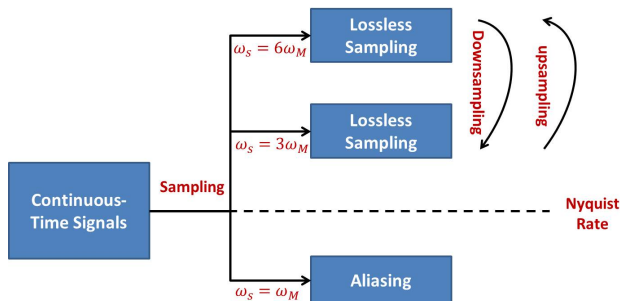


# Summary of DT Sampling

- Two operations: sampling vs. decimation
- What's their difference in frequency domain?
- $N\omega_M < \pi$  is always the criterion for lossless sampling



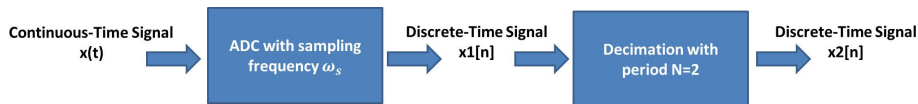
# Anything Else?



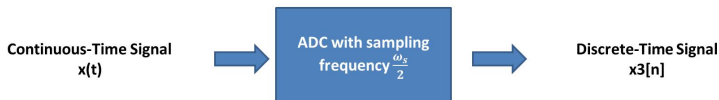
- **Downsampling:** to reduce the sampling frequency
- **Upsampling:** to generate a DT signal with higher sampling frequency
- As long as Nyquist rate is satisfied, the transform between low-sampling-frequency version and high-sampling-frequency versions is lossless.
- **How to do downsampling and upsampling in discrete-time domain?**

# Downsampling in DT Domain

- **Downsampling:** a general procedure to reduce the sampling frequency



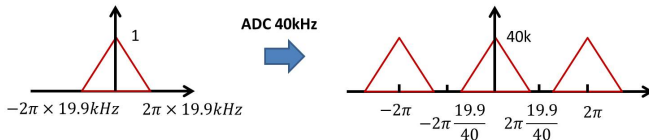
**Equivalent**



Obviously,  $x_2[n] = x_3[n]$   
Downsampling: from  $x_2[n]$  to  $x_3[n]$

## Downsampling Example (1/2)

- Suppose we have a clip of voice,  $x(t)$ , with bandwidth = 19.9kHz
- It can be converted to DT signal with sampling frequency 40kHz, denoted as  $x_1[n]$



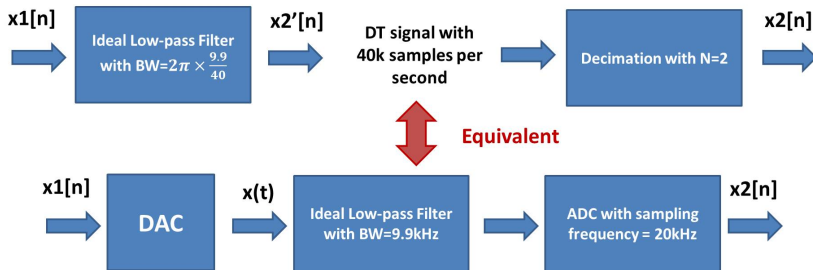
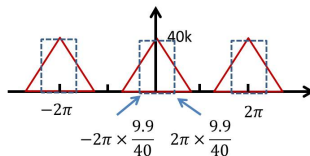
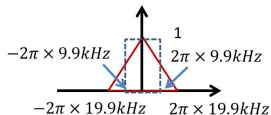
- Based on  $x_1[n]$ , if we want to save the voice information within 9.9kHz into another DT signal, what can we do?

**One Choice:**



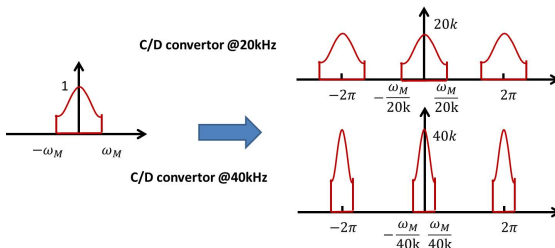
**How can we generate  $x_2[n]$  in discrete-time domain?**

# Downsampling Example (2/2)



# Upsampling

- **Upsampling**: a procedure to generate a sequence with higher sampling frequency
- Superpose the following two digital sound clips
  - ▶ Audio clip 1: Bandwidth= 19.9kHz, sampled at 40kHz
  - ▶ Audio clip 2: Bandwidth= 9.9kHz, sampled at 20kHz
- How to double the sampling frequency of audio clip 2 (40kHz), so that both clips can be merged?



# Recap: Time Reversal and Expansion

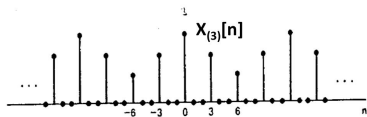
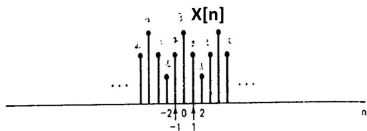
- Time Reversal

$$x[-n] \longleftrightarrow X(e^{-j\omega})$$

- Time Expansion

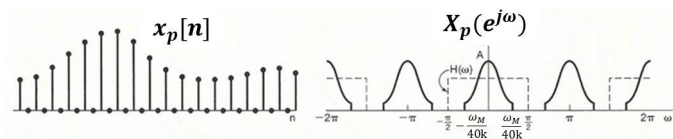
► Define  $x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n \text{ is a multiple of } k \\ 0, & \text{Otherwise} \end{cases}$ , then

$$x_{(k)}[n] \longleftrightarrow X(e^{jk\omega}) \quad (1)$$



- Time expansion (Insert zeros):

$$x_p[n] = x_{b(2)}[n] \longleftrightarrow X_p(e^{j\omega}) = X_b(e^{j2\omega})$$



- Low-pass filtering:

